

FULL DETAILED JAVASCRIPT NOTES WITH PRACTICE QUESTIONS & ANSWERS

1. BASICS OF JAVASCRIPT

JavaScript is a scripting language used in web development to add dynamic behavior to websites.

Topics:

- Variables (let, const, var)
- Data Types (string, number, boolean, undefined, null, object)
- Operators (+, -, *, /, %, ==, ===)

Example:

```
let x = 10;
```

```
let y = 20;
```

```
console.log(x + y);
```

Practice Question:

Q: What is the difference between == and === ?

A: == compares only values, === compares value + data type.

2. CONTROL FLOW

Control flow helps to decide conditions and repeat tasks.

```
if(condition) { }
```

```
else { }
```

Loops:

- for loop
- while loop
- do...while loop

Practice Question:

Q: Write a loop to print numbers 1 to 5.

A:

```
for(let i=1; i<=5; i++){  
  console.log(i);  
}
```

3. FUNCTIONS

Functions are blocks of code designed to perform tasks.

Example:

```
function add(a, b){  
  return a + b;  
}
```

Arrow Function:

```
const mul = (a, b) => a * b;
```

Practice Question:

Q: Write a function to return the square of a number.

A:

```
function square(n){  
  return n * n;  
}
```

4. ARRAYS & OBJECTS

Arrays store multiple values.

Example:

```
let arr = [1,2,3];
```

Array Methods:

- map()
- filter()
- reduce()

Objects store key-value pairs.

Example:

```
let user = {name: "Vishnu", age: 20};
```

Practice Question:

Q: Filter even numbers from [1,2,3,4,5].

A:

```
let even = [1,2,3,4,5].filter(n => n % 2 === 0);
```

5. DOM MANIPULATION

DOM allows JavaScript to interact with HTML.

```
document.getElementById("id")
```

```
document.querySelector(".class")
```

Events:

```
button.addEventListener("click", function(){});
```

Practice Question:

Q: Change text of a paragraph to "Hello".

A:

```
document.querySelector("p").textContent = "Hello";
```

6. ES6+ FEATURES

Includes:

- let, const
- Spread operator (...)
- Template literals `\${}`
- Destructuring

Example:

```
const arr = [1,2,3];  
const copy = [...arr];
```

Practice Question:

Q: What does the spread operator do?

A: Makes a copy of arrays/objects.

7. ASYNC JAVASCRIPT

Used for tasks like APIs.

Promises:

```
fetch(url).then(res => res.json());
```

Async/Await:

```
async function getData(){  
  let res = await fetch(url);  
}
```

Practice Question:

Q: What is async/await?

A: A cleaner way to write asynchronous code.

8. ADVANCED JS

Topics:

- Hoisting
- Closures
- Event Loop
- Prototypes
- this keyword

Practice Q:

Q: What is closure?

A: A function that remembers variables from its outer scope.

9. PRACTICE PROJECTS

-
- Calculator
 - Todo List
 - Weather App
 - Notes App

END OF DOCUMENT
